

Assignment - Unit I

1) There are three types of errors which are as follows:

(i) Absolute error: It is the numerical difference ~~between~~ between the true value of a quantity and its approximate value. approximates value

$$E_A = X - X_1, \text{ where } X \text{ is true value and } X_1 \text{ is } x$$

~~(ii)~~ (ii) Relative error: It is the numerical ^{division} difference between absolute error and true of a quantity i.e.

$$E_R = \frac{E_A}{X}$$

(iii) Percentage error: It is simply the relative expressed in terms of percentage error i.e. $E_p = 100 E_R$

2) It is simply the no. of figures that are known with some degree of reliability. Rules for deciding it are as follows:

(i) All non-zero digits are significant.

(ii) Zeros between non-zero digits are significant.

(iii) Zeros to the left of non-zero digits are not significant.

(iv) Zeros to the right of a decimal point in a no. are significant. the zeros are not necessarily significant.

(v) When a no. ends in zeros that are not to the right of a decimal.

3) Given the function $f(x) = x^3 + x^2 - 1$

Take $a = 0$,

$$f(0) = -1$$

Take $b = 1$,

$$f(1) = 1$$

$\therefore$ root of $f(x)$ exists between 0 and 1

a	b	f(a)	f(b)	x_i	$f(x_i)$
0	1	-1	1	0.5	-0.625
0.5	1	-0.625	1	0.75	-0.0156
0.75	1	-0.0156	1	0.875	0.4355
0.75	0.875	-0.0156	0.4355	0.8125	0.1965

$$0.75 \quad 0.8125 \quad -0.0156 \quad 0.1965 \quad 0.78125 \quad 0.0872$$

$$0.75 \quad 0.78125 \quad -0.0156 \quad 0.0872 \quad 0.765625 \quad 0.035$$

$$0.75 \quad 0.765625 \quad -0.0156 \quad 0.035 \quad 0.7578125 \quad 0.0095$$

$\therefore$ Root of $f(x) = x^3 + x^2 - 1$ is $x = 0.7578125$ (approx)

4/8) $f(x) = 4e^{-x} \sin x - 1$

Now,

At $x = 0$,

$$f(0) = -1$$

At $x = 1$,

$$f(1) = 0.2382$$

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

a	b	f(a)	f(b)	x_i	$f(x_i)$
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0	1	-1	0.2382	0.8076	0.289
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0	0.8076	-1	0.289	0.6266	0.2535
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0	0.6266	-1	0.2535	0.4999	0.1630
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0	0.4999	-1	0.1630	0.4298	0.0845
0	0.4298	-1	0.0845	0.3963	0.0388
0	0.3963	-1	0.0388	0.3815	0.0169
0	0.3815	-1	0.0169	0.3752	0.0072

∴ Solution to the equation $4e^{-x} \sin(x) - 1$ is $x = 0.3752$
(approx)

Q) Given that $f(x) = 5x \log x - 6$

differentiating both sides with respect to x

$$\Rightarrow f'(x) = 5(\log x + 1) \quad \cancel{= 5 \log x + 1}$$

Now, Take $x_0 = 2$

$$\therefore x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3.1934$$

$$x_2 = 2.6108$$

$$x_3 = 2.7922$$

$$x_4 = 2.7223$$

$$x_5 = 2.7475$$

$$x_6 = 2.7381$$

$$x_7 = 2.7416$$

$$x_8 = 2.7403$$

$$x_9 = 2.7408$$

$$x_0 = \leftarrow 2.7406$$

$$x_{11} = 2.7406$$

$\therefore$ Solution of the given equation is $x = 2.7406$

Q) Given the equation $x = \frac{1}{(1+x)^2}$

$$\Rightarrow \phi(x) = \frac{1}{(1+x)^2}$$

$$\Rightarrow \phi'(x) = \frac{1}{2(1+x)}$$

Now, for $x = 1.3$

$$\Rightarrow \frac{1}{2(1+1.3)} \Rightarrow \frac{1}{4.6} \Rightarrow 0.2174 < 1$$

for $x = 1.5$

$$\Rightarrow \frac{1}{2(1+1.5)} \Rightarrow \frac{1}{2(2.5)} = 0.2 < 1$$

$$\therefore x_{i+1} = \phi(x_i)$$

$\leftarrow$ Take $x_0 = 1.4$

$$\Rightarrow x_1 = \phi(x_0) = 0.5736$$

$$\Rightarrow x_2 = 0.7260$$

$$\Rightarrow x_3 = 0.3357$$

$$x_4 = 0.5605$$

$$x_5 = 0.4106$$

$$x_6 = 0.5025$$

$$x_7 = 0.4429$$

$$x_8 = 0.4803$$

$$x_9 = 0.4564$$

$$x_{10} = 0.4715$$

$$x_{11} = 0.4618$$

$$x_{12} = 0.4680$$

$$x_{13} = 0.46$$

$\therefore$ Root of the given equation is $x = 0.4803$ (approx)

Q) Given the equation

$$f(x) = x \log x - 1.2$$

Initial approximations, $x_0 = 2$, $x_1 = 3$

$$\therefore x_{i+2} = \frac{x_i f(x_{i+1}) - x_{i+1} f(x_i)}{f(x_{i+1}) - f(x_i)}$$

$$\therefore x_2 = 2.721$$

$$x_3 = 2.7402$$

$$x_4 = 2.7406$$

$$x_5 = 2.7406$$

~~Ans~~

∴ Root of the given equation $x \log x = 1.2$ is
 $x = 2.7406$ correct upto 4 decimal places.